



PHL 1010: Magnetostatics



Many historians of science believe that the compass, which uses a magnetic needle, was used in China as early as the 13th century BC, **its invention being of Indian origin...**

Raymond A. Serway

John W. Jewett

Loadstone



Thales of Miletus



1819: Current creates magnetic field
...at last determined relation between E & M

HANS CHRISTIAN OERSTED

Danish Physicist and Chemist (1777–1851)



1831: Law of EM induction, Self/mutual inductance,
Magneto-optic effect...

MICHAEL FARADAY

British Physicist and Chemist (1791–1867)

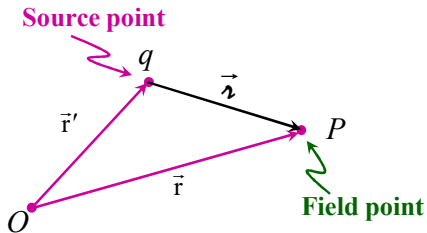


1864: Vacuum polarization current (electric
counterpart of Faraday's magnetic induction)
EM Theory (field equations)

JAMES CLERK MAXWELL

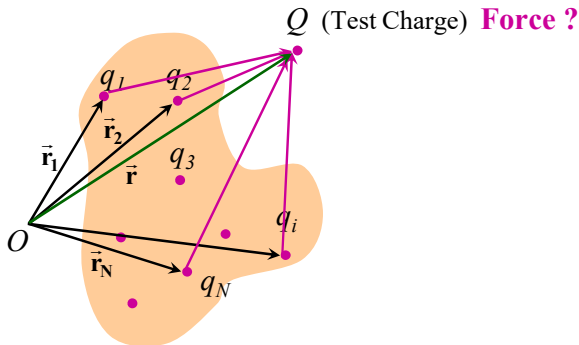
Scottish Theoretical Physicist (1831–1879)

Convention, that would be followed...

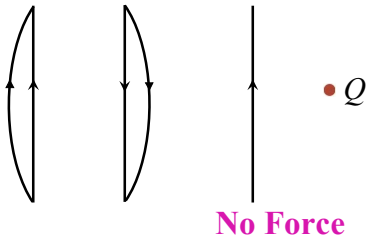


$$\vec{\mathbf{z}} = \vec{r} - \vec{r}'$$

The Fundamental Problem of EM Theory

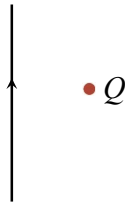
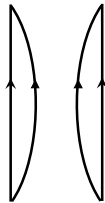


Consider the simplest & Historical Experiment





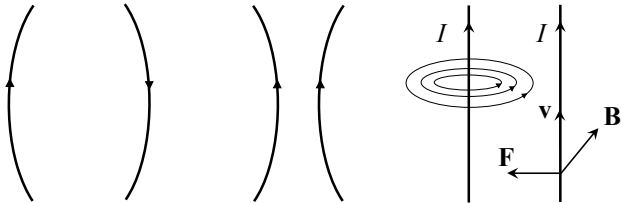
Consider the simplest & Historical Experiment



No Force



Consider the simplest & Historical Experiment



Magnetic Force

$$\text{Lorentz Force Law: } \vec{F}_{\text{mag}} = q(\vec{v} \times \vec{B})$$

$$\vec{F}_{\text{tot}} = q[\vec{E} + (\vec{v} \times \vec{B})]$$

Work done by magnetic force: q moves by $d\vec{l} = \vec{v} dt$

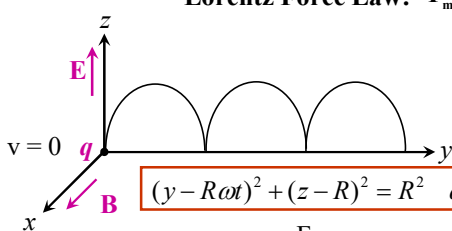
$$dW_{\text{mag}} = \vec{F}_{\text{mag}} \cdot d\vec{l} = Q(\vec{v} \times \vec{B}) \cdot \vec{v} dt = 0 \quad \text{Magnetic force does no work}$$

Magnetic Force can never accelerate/decelerate a particle set in motion

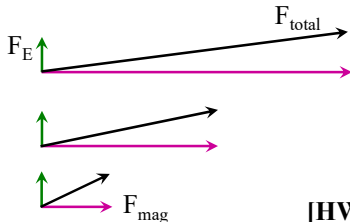
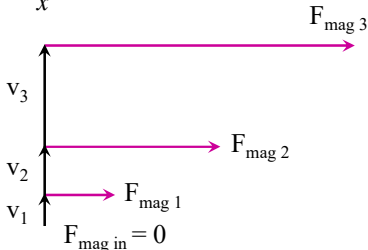
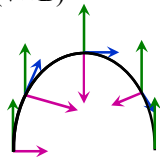
[Example 5.3 HW]

Motion of Charges particle under Magnetic Force

Lorentz Force Law: $\vec{F}_{\text{mag}} = q(\vec{v} \times \vec{B})$



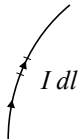
$$(y - R\omega t)^2 + (z - R)^2 = R^2 \quad \omega = qB/m; \quad R = E/\omega B$$



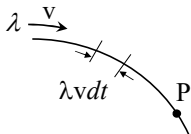
[HW]

Current: $I = \frac{dq}{dt}$

Line current



$$dI = I dl \quad \checkmark$$



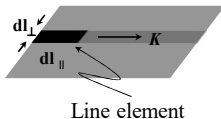
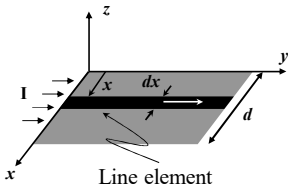
$$I = \frac{dq}{dt} = \frac{\lambda v dt}{dt} = \lambda v$$

$$\vec{I} = \lambda \vec{v}$$

Line current distribution $\longrightarrow$ Line current density

Important concepts

Surface Current



$$dI = \left(\frac{I}{d} \right) dx = K dx$$

$$I = \int \vec{K} \cdot d\vec{l}$$

$$K = \frac{dI}{dx}$$

$$dI = \frac{dQ}{dt} = \frac{\sigma dl_{\perp} dl_{\parallel}}{dt} = \sigma dl_{\perp} \frac{dl_{\parallel}}{dt} = \sigma dl_{\perp} v$$

$$K = \frac{dI}{dl_{\perp}}$$

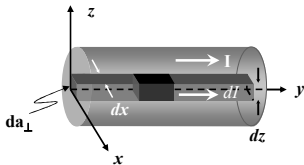
$$\frac{dI}{dl_{\perp}} = \sigma v = K$$

$$\sigma \vec{v} = \vec{K}$$

Surface current distribution $\longrightarrow$ Surface current density

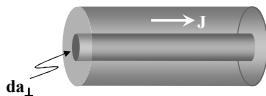
Important concepts

Volume Current



$$dI = \left(\frac{I}{\pi R^2} \right) dx dz$$

$$J = \frac{dI}{dx dz}$$



$$J = \frac{dI}{da_{\perp}} \quad I = \int \vec{J} \cdot d\vec{a}$$

$$dI = \frac{dQ}{dt} = \frac{\rho da_{\perp} dy}{dt}$$

$$= \rho da_{\perp} \frac{dy}{dt} = \rho da_{\perp} v$$

$$\frac{dI}{da_{\perp}} = \rho v = J \quad \rho \vec{v} = \vec{J}$$

Volume current distribution $\longrightarrow$ Volume current density

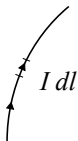
Example: Current density is preoperational to the distance from the axis $J = k r$
Find total current.



$$I = \int \vec{\mathbf{J}} \cdot d\vec{\mathbf{a}}$$

$$I = \int (kr) r dr d\phi = 2\pi k \int_0^a r^2 dr = \frac{2\pi k a^3}{3} \quad \checkmark$$

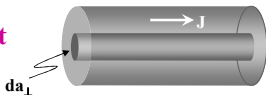
Line current



Surface current



Volume current



$$dI = I dl = K dx = J dx dz$$

Consequence of Lorentz Force: Force on Current

Magnetic Force on a current element:

$$\vec{F}_{\text{mag}} = \int dq (\vec{v} \times \vec{B}) = \int \lambda dl (\vec{v} \times \vec{B}) = \int dl (\vec{I} \times \vec{B}) = I \int (d\vec{l} \times \vec{B})$$

Magnetic force on surface current density:

$$\vec{F}_{\text{mag}} = \int dq (\vec{v} \times \vec{B}) = \int \sigma da (\vec{v} \times \vec{B}) = \int (\vec{K} \times \vec{B}) da$$

Magnetic force on volume current density:

$$\vec{F}_{\text{mag}} = \int dq (\vec{v} \times \vec{B}) = \int \rho d\tau (\vec{v} \times \vec{B}) = \int (\vec{J} \times \vec{B}) d\tau$$

Important concepts

$I = \oint \vec{J} \cdot d\vec{a}$ = total charge passing through a closed surface per unit time

$$= \int_V (\vec{\nabla} \cdot \vec{J}) d\tau$$

= total charge leaving a volume V per unit time

= rate of which charges are decreasing with time within the closed surface

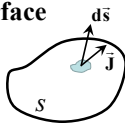
$$= -\frac{\partial}{\partial t} \int_V \rho d\tau = -\int_V \frac{\partial \rho}{\partial t} d\tau \longrightarrow \int_V \left(\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t} \right) d\tau = 0$$

$$\longrightarrow \boxed{\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0}$$

Continuity Equation

Conservation of charge

Imp: $\frac{\partial \rho}{\partial t} = 0 \longrightarrow$ **Steady state current** $\longrightarrow \vec{\nabla} \cdot \vec{J} = 0$



Magnetic Field of a Steady Current

Stationary charges $\longrightarrow$ Constant electric field **Electrostatics**

Steady current $\longrightarrow$ Constant magnetic field **Magnetostatics**

$$\frac{\partial \rho}{\partial t} = 0; \quad \vec{\nabla} \cdot \vec{J} = 0$$

Magnetic field of a steady current: Biot-Savart Law

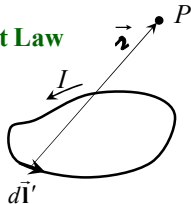
$$\vec{B}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \int \frac{d\vec{l}' \times \hat{\mathbf{r}}}{r^2}$$

$$\vec{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{K}(\mathbf{r}') \times \hat{\mathbf{r}}}{r^2} da'$$

$$\vec{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}') \times \hat{\mathbf{r}}}{r^2} d\tau' \quad \mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$$

$$1 \text{ T} = 1 \text{ N/A.m}$$

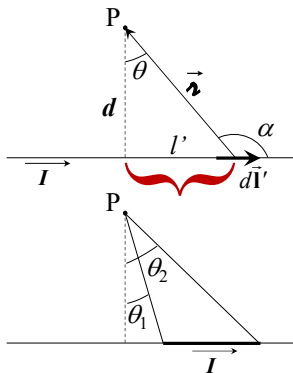
Principal of superposition applicable to magnetic fields



Magnetic Field of a Steady Current

Magnetic field of a steady current: Biot-Savart Law

Magnetic field at a distance d from a wire carrying steady state current I



$$\vec{B}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \int \frac{d\vec{l}' \times \hat{\mathbf{r}}}{r^2} \quad 90^\circ - \theta + \alpha = 180^\circ$$

$$|d\vec{l}' \times \hat{\mathbf{r}}| = dl' \sin \alpha = dl' \cos \theta$$

$$\tan \theta = \frac{l'}{d} \rightarrow dl' = d \sec^2 \theta d\theta$$

$$\cos \theta = \frac{d}{r} \rightarrow \frac{1}{r^2} = \frac{\cos^2 \theta}{d^2}$$

$$B = \frac{\mu_0 I}{4\pi} \int_{\theta_1}^{\theta_2} d \sec^2 \theta d\theta \cos \theta \frac{\cos^2 \theta}{d^2}$$

$$= \frac{\mu_0 I}{4\pi d} \int_{\theta_1}^{\theta_2} d\theta \cos \theta = \frac{\mu_0 I}{4\pi d} (\sin \theta_2 - \sin \theta_1)$$

Magnetic Field of a Steady Current

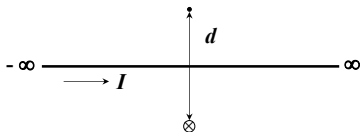
Magnetic field of a steady current: Biot-Savart Law

Magnetic field at a distance d from a wire carrying steady state current I

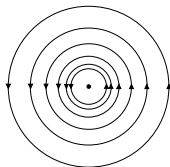
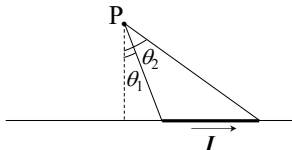
$$B = \frac{\mu_0 I}{4\pi d} (\sin \theta_2 - \sin \theta_1)$$

In case of infinite current carrying wire

$$\theta_2 \rightarrow 90^\circ; \theta_1 \rightarrow -90^\circ$$



$$B = \frac{\mu_0 I}{2\pi d}$$



$$\vec{B} = \frac{\mu_0 I}{2\pi d} \hat{\phi}$$

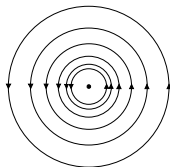
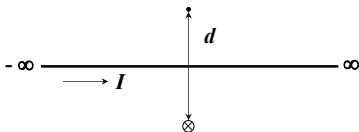
Magnetic Field of a Steady Current

Magnetic field of a steady current: Biot-Savart Law

Magnetic field at a distance d from a wire carrying steady state current I

$$B = \frac{\mu_0 I}{4\pi d} (\sin \theta_2 - \sin \theta_1)$$

In case of infinite current carrying wire



Magnetic field has finite curl !
something you will never see in an electrostatic field

Divergence & Curl of Magnetic Field

General form of Biot-Savart law :

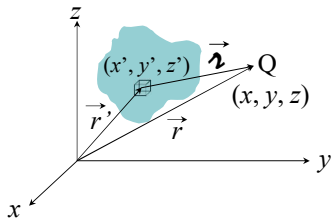
$$\vec{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\mathbf{r}') \times \hat{\mathbf{r}}}{r^2} d\tau'$$

$$\vec{B} = \vec{B}(x, y, z)$$

$$\vec{J} = \vec{J}(x', y', z')$$

$$r^2 = (x - x')^2 + (y - y')^2 + (z - z')^2$$

$$d\tau' = dx' dy' dz'$$



Integration is over primed coordinates. Let's take div and curl w. r. t. unprimed coordinates.

$$\vec{\nabla} \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\vec{\nabla} \times \vec{A}) - \vec{A} \cdot (\vec{\nabla} \times \vec{B})$$

$$\vec{\nabla} \cdot \vec{B} = \frac{\mu_0}{4\pi} \int \vec{\nabla} \cdot \left(\vec{J} \times \frac{\hat{\mathbf{r}}}{r^2} \right) d\tau'$$

$$\vec{\nabla} \cdot \left(\vec{J} \times \frac{\hat{\mathbf{r}}}{r^2} \right) = \frac{\hat{\mathbf{r}}}{r^2} \cdot (\vec{\nabla} \times \vec{J}) - \vec{J} \cdot \left(\vec{\nabla} \times \frac{\hat{\mathbf{r}}}{r^2} \right) = 0 \rightarrow \boxed{\vec{\nabla} \cdot \vec{B} = 0}$$

Divergence & Curl of Magnetic Field

General form of Biot-Savart law :

$$\vec{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\mathbf{r}') \times \hat{\mathbf{r}}}{r^2} d\tau'$$

$$\vec{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\mathbf{r}') \times \vec{r}}{r^3} d\tau'$$

But $\frac{\vec{r}}{r^3} = -\nabla \left(\frac{1}{r} \right)$

$$\vec{\nabla} \times (f\vec{A}) = f \vec{\nabla} \times \vec{A} - \vec{A} \times \vec{\nabla} f$$

$$\vec{B}(\mathbf{r}) = -\frac{\mu_0}{4\pi} \int \vec{J}(\mathbf{r}') \times \nabla \left(\frac{1}{r} \right) d\tau'$$

$$\vec{\nabla} \times \frac{\vec{J}(\mathbf{r}')}{r} = \frac{1}{r} \cancel{\vec{\nabla} \times \vec{J}(\mathbf{r}')} - \vec{J}(\mathbf{r}') \times \vec{\nabla} \frac{1}{r}$$

$$\therefore \vec{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \vec{\nabla} \times \frac{\vec{J}(\mathbf{r}')}{r} d\tau'$$

$$\vec{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \vec{\nabla} \times \int \frac{\vec{J}(\mathbf{r}')}{r} d\tau'$$

Divergence & Curl of Magnetic Field

$$\therefore \vec{\mathbf{B}}(\mathbf{r}) = \frac{\mu_0}{4\pi} \vec{\nabla} \times \int \frac{\vec{\mathbf{J}}(\mathbf{r}')}{r} d\tau'$$

Taking curl $\vec{\nabla} \times \vec{\mathbf{B}}(\mathbf{r}) = \frac{\mu_0}{4\pi} \vec{\nabla} \times \vec{\nabla} \times \int \frac{\vec{\mathbf{J}}(\mathbf{r}')}{r} d\tau'$

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{\mathbf{A}}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{\mathbf{A}}) - \nabla^2 \vec{\mathbf{A}}$$

$$\vec{\nabla} \cdot (f\vec{\mathbf{A}}) = f \vec{\nabla} \cdot \vec{\mathbf{A}} + \vec{\nabla} f \cdot \vec{\mathbf{A}}$$

$$\vec{\nabla} \times \vec{\nabla} \times \int \frac{\vec{\mathbf{J}}(\mathbf{r}')}{r} d\tau' = \vec{\nabla}(\vec{\nabla} \cdot \int \frac{\vec{\mathbf{J}}(\mathbf{r}')}{r} d\tau') - \nabla^2 \int \frac{\vec{\mathbf{J}}(\mathbf{r}')}{r} d\tau'$$

$$= \vec{\nabla} \left(\int \vec{\nabla} \cdot \frac{\vec{\mathbf{J}}(\mathbf{r}')}{r} d\tau' \right) - \int \nabla^2 \frac{\vec{\mathbf{J}}(\mathbf{r}')}{r} d\tau'$$

$$= \vec{\nabla} \int \left[\cancel{\frac{1}{r} \vec{\nabla} \cdot \vec{\mathbf{J}}(\mathbf{r}')} + \vec{\mathbf{J}}(\mathbf{r}') \cdot \vec{\nabla} \left(\frac{1}{r} \right) \right] d\tau' - \int \vec{\mathbf{J}}(\mathbf{r}') \nabla^2 \left(\frac{1}{r} \right) d\tau'$$

$$\vec{\nabla} \times \vec{\mathbf{B}}(\mathbf{r}) = \frac{\mu_0}{4\pi} \vec{\nabla} \int \vec{\mathbf{J}}(\mathbf{r}') \cdot \vec{\nabla} \left(\frac{1}{r} \right) d\tau' - \frac{\mu_0}{4\pi} \int \vec{\mathbf{J}}(\mathbf{r}') \nabla^2 \left(\frac{1}{r} \right) d\tau'$$

Divergence & Curl of Magnetic Field

$$\vec{\nabla} \times \vec{\mathbf{B}}(\mathbf{r}) = \frac{\mu_0}{4\pi} \vec{\nabla} \int \vec{\mathbf{J}}(\mathbf{r}') \cdot \vec{\nabla} \left(\frac{1}{r} \right) d\tau' - \frac{\mu_0}{4\pi} \int \vec{\mathbf{J}}(\mathbf{r}') \nabla^2 \left(\frac{1}{r} \right) d\tau$$

$$\vec{\nabla} \left(\frac{1}{r} \right) = -\vec{\nabla}' \left(\frac{1}{r} \right) \quad \nabla^2 \left(\frac{1}{r} \right) = -4\pi \delta^3(\vec{r})$$

$$\vec{\nabla} \times \vec{\mathbf{B}}(\mathbf{r}) = -\frac{\mu_0}{4\pi} \vec{\nabla} \int \vec{\mathbf{J}}(\mathbf{r}') \cdot \vec{\nabla}' \left(\frac{1}{r} \right) d\tau' + \frac{\mu_0}{4\pi} \int \vec{\mathbf{J}}(\mathbf{r}') 4\pi \delta^3(\vec{r}) d\tau$$

Integration by part yields

$$\vec{\nabla} \times \vec{\mathbf{B}}(\mathbf{r}) = \frac{\mu_0}{4\pi} \vec{\nabla} \int \left(\frac{\vec{\nabla}' \cdot \vec{\mathbf{J}}(\mathbf{r}')}{r} \right) d\tau' - \frac{\mu_0}{4\pi} \oint_s \frac{\vec{\mathbf{J}}(\mathbf{r}')}{r} da' + \mu_0 \vec{\mathbf{J}}(\mathbf{r})$$

$$\vec{\nabla} \times \vec{\mathbf{B}}(\mathbf{r}) = \mu_0 \vec{\mathbf{J}}(\mathbf{r})$$

Ampere's Law

Ampere's Law

$$\vec{\nabla} \times \vec{\mathbf{B}} = \mu_0 \vec{\mathbf{J}}$$

$$\int_S (\vec{\nabla} \times \vec{\mathbf{B}}) \cdot d\vec{\mathbf{a}} = \mu_0 \int_S \vec{\mathbf{J}} \cdot d\vec{\mathbf{a}}$$

$$\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{l}} = \mu_0 I_{\text{en}}$$

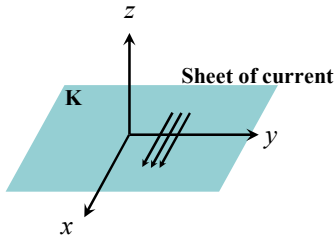
Right hand rule for direction of integration and + ve current



**Application demand
high degree of symmetry**

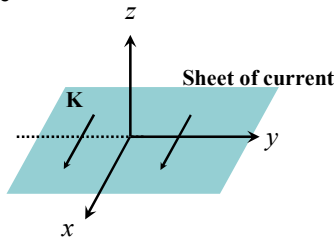
Ampere's Law – Applications (2)

Field of an infinite uniform surface current $\mathbf{K} = K \hat{\mathbf{x}}$ flowing over xy-plane



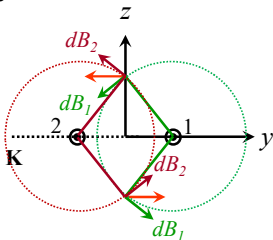
Ampere's Law – Applications (2)

Field of an infinite uniform surface current $\mathbf{K} = K \hat{\mathbf{x}}$ flowing over xy-plane



Ampere's Law – Applications (2)

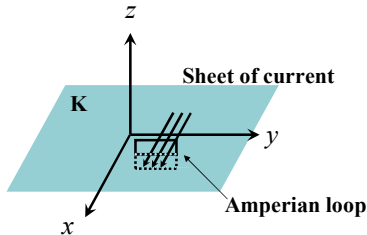
Field of an infinite uniform surface current $\mathbf{K} = K \hat{x}$ flowing over xy-plane



→ Field above is towards left, and field below is towards right

Ampere's Law – Applications (2)

Field of an infinite uniform surface current $\mathbf{K} = K \hat{\mathbf{x}}$ flowing over xy-plane



$$\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{l}} = 2Bl = \mu_0 I_{\text{en}} = \mu_0 K l$$

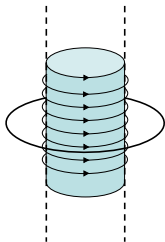
$$\vec{\mathbf{B}} = -\mu_0 K \hat{\mathbf{y}} / 2 \quad z > 0$$

$$\vec{\mathbf{B}} = \mu_0 K \hat{\mathbf{y}} / 2 \quad z < 0$$

[HW:
Examples: 5.9 & 5.10]

Ampere's Law – Applications (3)

Field of very long solenoid (n closely wound turn per unit length)



Could field have radial component ?

NO

Could field have circumferential component ?

NO

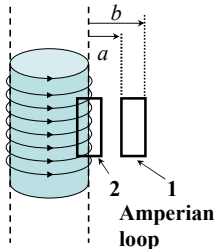
$$\oint \vec{B} \cdot d\vec{l} = B_{\phi}(2\pi r) = \mu_0 I_{\text{en}} = 0$$



Field runs parallel to the axis!

Ampere's Law – Applications (3)

Field of very long solenoid (n closely wound turn per unit length)



Field runs parallel to the axis!

Loop 1:

$$\oint \vec{B} \cdot d\vec{l} = [B(a) - B(b)] L = \mu_0 I_{\text{en}} = 0$$

$$B(a) = B(b)$$

Loop 2:

$$\oint \vec{B} \cdot d\vec{l} = B L = \mu_0 I_{\text{en}} = \mu_0 n I L$$

$$\mathbf{B}_{\text{in}} = \mu_0 n I \hat{\mathbf{z}}$$

$$\mathbf{B}_{\text{out}} = 0$$

HE: Example 5.10

So far.....

Maxwell Equation for Electrostatics & Magnetostatics

$$\vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0 \quad \text{Gauss's law}$$

$$\vec{\nabla} \times \vec{E} = 0 \quad \text{No name}$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad \text{No name}$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \mathbf{J} \quad \text{Ampere's Law}$$

→ Magnetic monopole does not exist!

Electric forces are enormously larger than magnetic one

Velocity of source & test charge —————→ Velocity of light

Magnetic field strength —————→ Electric field strength

Magnetic Vector Potential

In Magnetostatics $\vec{\nabla} \times \vec{\mathbf{B}} = \mu_0 \mathbf{J}$ $\vec{\nabla} \cdot \vec{\mathbf{B}} = 0$

In Electrostatics $\vec{\nabla} \times \vec{\mathbf{E}} = 0 \longrightarrow \vec{\mathbf{E}} = -\vec{\nabla} V$

**Calculation of
Electric Field**

**Calculation of
Electric Potential**

$\vec{\nabla} \times \vec{\mathbf{B}} \neq 0 \longrightarrow \vec{\mathbf{B}} \neq -\vec{\nabla} \phi$

Strategy $\vec{\nabla} \cdot \vec{\mathbf{B}} = 0 \longrightarrow \mathbf{B}$ can be expressed as a curl of a vector (magnetostatic) potential

$$\vec{\mathbf{B}} = \vec{\nabla} \times \vec{\mathbf{A}}$$

$$\vec{\mathbf{B}}(\mathbf{r}) = \frac{\mu_0}{4\pi} \vec{\nabla} \times \int \frac{\mathbf{J}(\mathbf{r}')}{r} d\tau' \longrightarrow \vec{\mathbf{A}}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}')}{r} d\tau'$$

Magnetic Vector Potential

$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int_V \frac{\mathbf{J}(\mathbf{r}')}{r} d\tau'$$

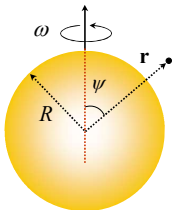
$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int_V \frac{\mathbf{I}}{r} dl' = \frac{\mu_0}{4\pi} \int I \frac{d\mathbf{l}'}{r}$$

$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int_V \frac{\mathbf{K}}{r} da'$$

Calculation of magnetic Field $\longrightarrow$ Calculation of magnetic Potential

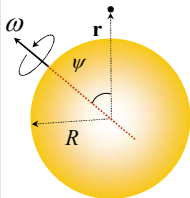
Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



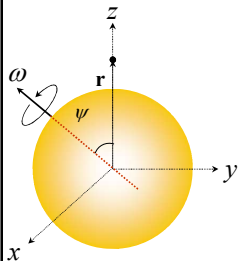
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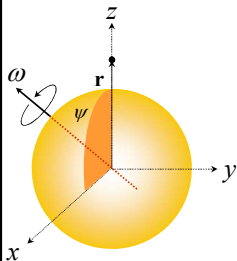
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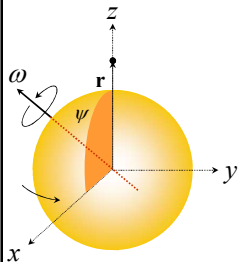
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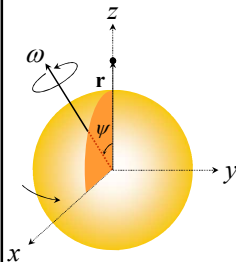
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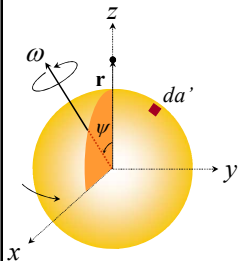
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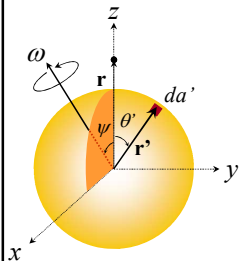
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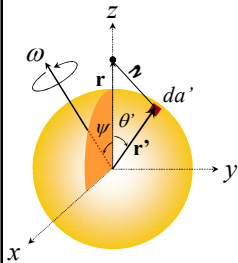
Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



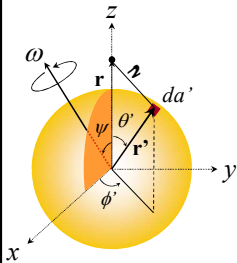
Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



Example - 1

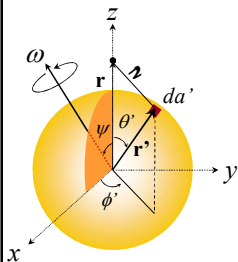
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Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

Surface charge $\longrightarrow$ Surface current

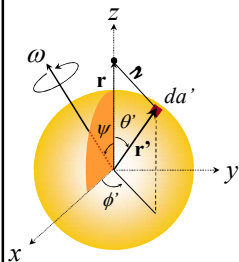


Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

Surface charge $\longrightarrow$ Surface current

$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{K}}{r} da'$$



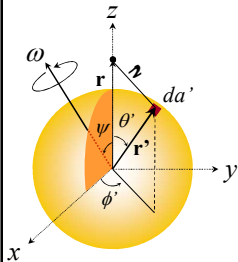
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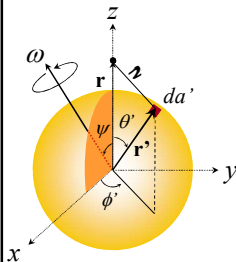
$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{K}}{r} da'$$

$$\vec{K} = \sigma \vec{v}$$



Example - 1

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Surface charge $\longrightarrow$ Surface current

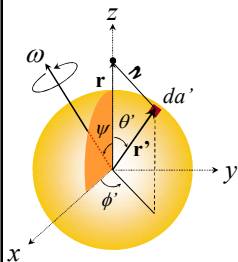
$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{K}}{\mathcal{r}} da'$$

$$\vec{K} = \sigma \vec{v}$$

$$\mathcal{r} = \sqrt{R^2 + r^2 - 2Rr \cos \theta'}$$

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



Surface charge $\longrightarrow$ Surface current

$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{K}}{z} da'$$

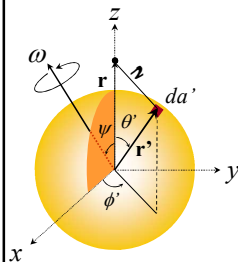
$$\vec{K} = \sigma \vec{v}$$

$$z = \sqrt{R^2 + r^2 - 2Rr \cos \theta'}$$

$$da' = R^2 \sin \theta' d\theta' d\phi'$$

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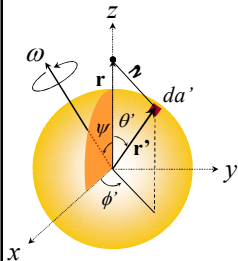
$$r = \sqrt{R^2 + r^2 - 2Rr \cos\theta'}$$

$$da' = R^2 \sin\theta' d\theta' d\phi'$$

$$\vec{A}(\vec{r}) =$$

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



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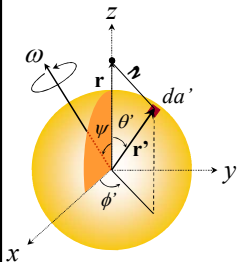
$$da' = R^2 \sin \theta' d\theta' d\phi'$$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int_0^{2\pi} \int_0^\pi$$

$$R^2 \sin \theta' d\theta' d\phi'$$

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



Surface charge $\longrightarrow$ Surface current

$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{K}}{\rho} da'$$

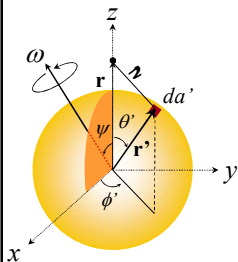
$$\vec{K} = \sigma \vec{v}$$

$$\rho = \sqrt{R^2 + r^2 - 2Rr \cos \theta'} \quad da' = R^2 \sin \theta' d\theta' d\phi'$$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int_0^{2\pi} \int_0^\pi \frac{\sigma \vec{\omega} \times \vec{r}'}{\rho} R^2 \sin \theta' d\theta' d\phi'$$

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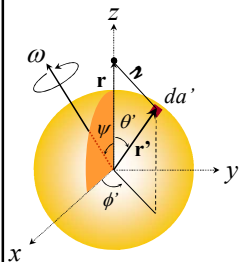
$$\vec{K} = \sigma \vec{v}$$

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Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



$$\vec{v} = \vec{\omega} \times \vec{r}$$

Surface charge $\longrightarrow$ Surface current

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{K}}{z} da'$$

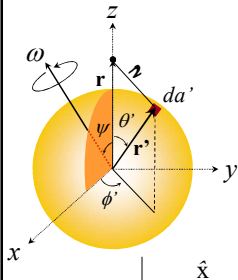
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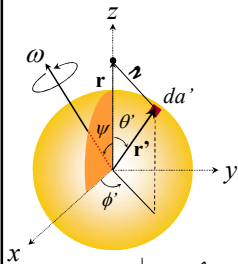
$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int_0^{2\pi} \int_0^\pi \frac{\sigma \vec{\omega} \times \vec{r}'}{\sqrt{R^2 + r^2 - 2Rr \cos \theta'}} R^2 \sin \theta' d\theta' d\phi'$$

$\hat{y}$

$\hat{z}$

Example - 1

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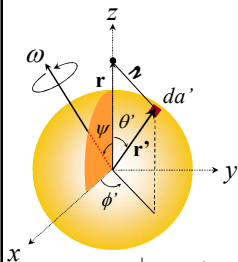
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$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int_0^{2\pi} \int_0^\pi \frac{\sigma \vec{\omega} \times \vec{r}'}{\sqrt{R^2 + r^2 - 2Rr \cos \theta'}} R^2 \sin \theta' d\theta' d\phi'$$

$$\vec{v} = \vec{\omega} \times \vec{r} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \omega \sin \psi & 0 & \omega \cos \psi \end{vmatrix}$$

Example - 1

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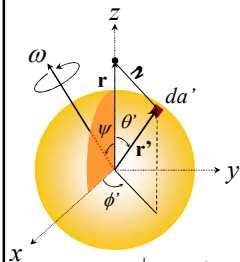
$$da' = R^2 \sin \theta' d\theta' d\phi'$$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int_0^{2\pi} \int_0^\pi \frac{\sigma \vec{\omega} \times \vec{r}'}{\sqrt{R^2 + r^2 - 2Rr \cos \theta'}} R^2 \sin \theta' d\theta' d\phi'$$

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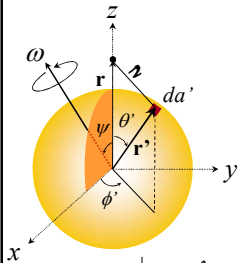
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$$= R\omega [-(\cos \psi \sin \theta' \sin \phi') \hat{x} + \dots]$$

Example - 1

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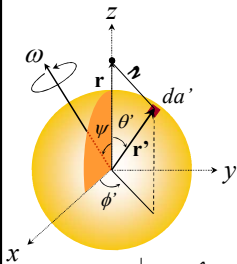
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$$= R\omega [-(\cos \psi \sin \theta' \sin \phi')\hat{x} + (\cos \psi \sin \theta' \cos \phi' - \sin \psi \cos \theta')\hat{y}]$$

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



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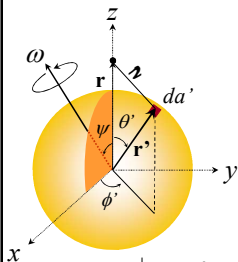
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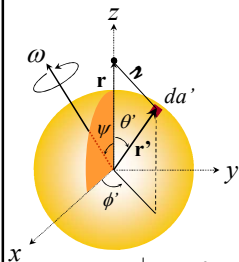
$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int_0^{2\pi} \int_0^\pi \frac{\sigma \vec{\omega} \times \vec{r}'}{\sqrt{R^2 + r^2 - 2Rr \cos \theta'}} R^2 \sin \theta' d\theta' d\phi'$$

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$$= R\omega [-(\cos \psi \sin \theta' \sin \phi')\hat{x} + (\cos \psi \sin \theta' \cos \phi' - \sin \psi \cos \theta')\hat{y} + (\sin \psi \sin \theta' \sin \phi')\hat{z}]$$

Example - 1

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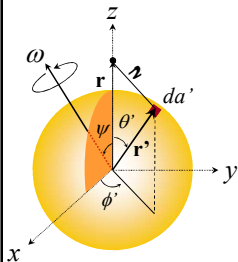
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$$= R\omega [-(\cos \psi \sin \theta' \sin \phi') \hat{x} + (\cos \psi \sin \theta' \cos \phi' - \sin \psi \cos \theta') \hat{y} + (\sin \psi \sin \theta' \sin \phi') \hat{z}]$$

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?

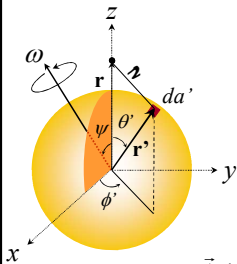


$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int_0^{2\pi} \int_0^\pi \frac{\sigma \vec{\omega} \times \vec{r}'}{\sqrt{R^2 + r^2 - 2Rr \cos \theta'}} R^2 \sin \theta' d\theta' d\phi'$$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int_0^{2\pi} \int_0^\pi \frac{\sigma R \omega [-\sin \psi \cos \theta'] \hat{y}}{\sqrt{R^2 + r^2 - 2Rr \cos \theta'}} R^2 \sin \theta' d\theta' d\phi'$$

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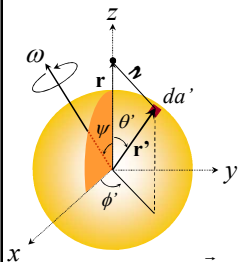
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Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



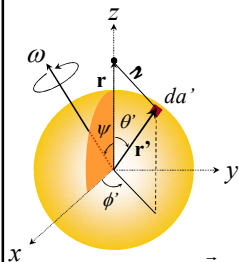
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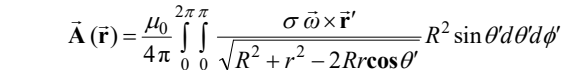


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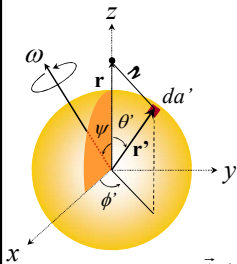


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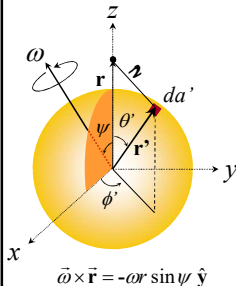
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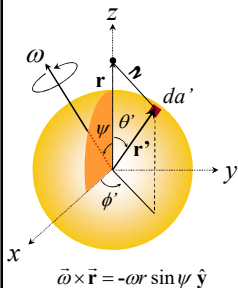
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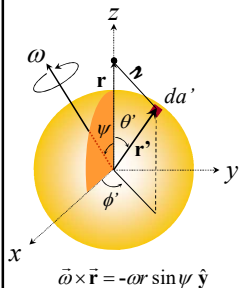


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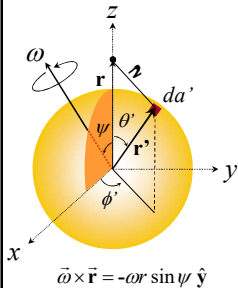


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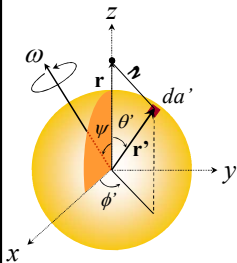
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No reference of the coordinate system – General Solution

Example - 1

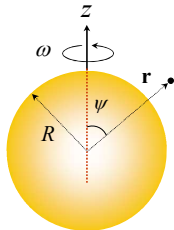
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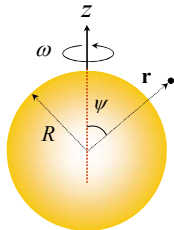


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Returning back to the original configuration, where ω coincides with the z -axis

Example - 1

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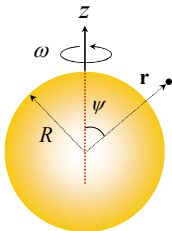
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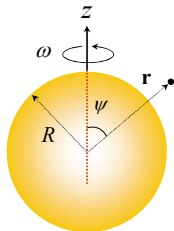
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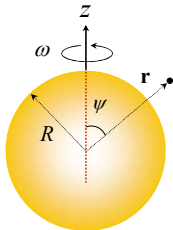
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Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



Inside the sphere

$\vec{B}$

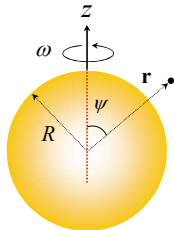
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Returning back to the original configuration, where ω coincides with the z -axis

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Example - 1

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Inside the sphere

$$\vec{B} = \vec{\nabla} \times \vec{A}$$

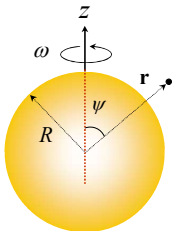
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Returning back to the original configuration, where ω coincides with the z-axis

$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R \omega}{3} r \sin \psi \hat{\phi} & \text{inside } (R > r) \\ \frac{\mu_0 \sigma R^4 \omega}{3r^2} \sin \psi \hat{\phi} & \text{outside } (R < r) \end{cases}$$

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R}{3} (\vec{\omega} \times \vec{r}) & \text{inside } (R > r) \\ \frac{\mu_0 \sigma R^4}{3r^3} (\vec{\omega} \times \vec{r}) & \text{outside } (R < r) \end{cases}$$

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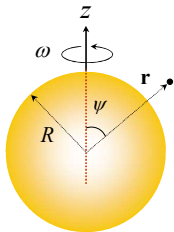
$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R \omega}{3} r \sin \psi \hat{\phi} & \text{inside } (R > r) \\ \frac{\mu_0 \sigma R^4 \omega}{3r^2} \sin \psi \hat{\phi} & \text{outside } (R < r) \end{cases}$$

Inside the sphere

$$\vec{B} = \vec{\nabla} \times \vec{A} = \frac{2\mu_0 \sigma R \omega}{3} (\cos \theta \hat{r} - \sin \theta \hat{\theta}) :$$

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



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Returning back to the original configuration, where ω coincides with the z -axis

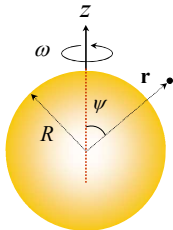
$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R \omega}{3} r \sin \psi \hat{\phi} & \text{inside } (R > r) \\ \frac{\mu_0 \sigma R^4 \omega}{3r^2} \sin \psi \hat{\phi} & \text{outside } (R < r) \end{cases}$$

Inside the sphere

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Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



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Returning back to the original configuration, where ω coincides with the z -axis

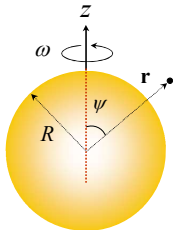
$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R \omega}{3} r \sin \psi \hat{\phi} & \text{inside } (R > r) \\ \frac{\mu_0 \sigma R^4 \omega}{3r^2} \sin \psi \hat{\phi} & \text{outside } (R < r) \end{cases}$$

Inside the sphere

$$\vec{B} = \vec{\nabla} \times \vec{A} = \frac{2\mu_0 \sigma R \omega}{3} (\cos \theta \hat{r} - \sin \theta \hat{\theta}) = \frac{2}{3} \mu_0 \sigma R \omega \hat{z} = \frac{2}{3} \mu_0 \sigma R \vec{\omega}$$

Example - 1

Spherical shell of radius R , carrying uniform surface charge σ is spinning at angular velocity ω . Vector potential at $\vec{r}$?



$$\vec{A}(\vec{r}) = \begin{cases} \frac{\mu_0 \sigma R}{3} (\vec{\omega} \times \vec{r}) & \text{inside } (R > r) \\ \frac{\mu_0 \sigma R^4}{3r^3} (\vec{\omega} \times \vec{r}) & \text{outside } (R < r) \end{cases}$$

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Inside the sphere

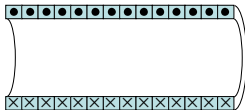
$$\vec{B} = \vec{\nabla} \times \vec{A} = \frac{2\mu_0 \sigma R \omega}{3} (\cos \theta \hat{r} - \sin \theta \hat{\theta}) = \frac{2}{3} \mu_0 \sigma R \omega \hat{z} = \frac{2}{3} \mu_0 \sigma R \vec{\omega}$$

Field is Uniform inside

Important concepts - 2

Magnetic Vector Potential

Vector potential of an infinite solenoid, radius R with n turns per unit length and current I ?



We can not use $\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}')}{r} d\tau'$ $\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{I}}{r} dl'$

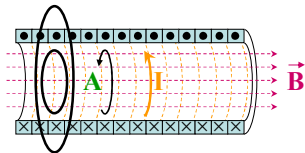
$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{K}}{r} da'$$

Cause current extends to infinity !

Important concepts - 2

Magnetic Vector Potential

Vector potential of an infinite solenoid, radius R with n turns per unit length and current I ?



$$\oint \vec{A} \cdot d\vec{l} = \int (\vec{\nabla} \times \vec{A}) \cdot d\vec{a} = \int \vec{B} \cdot d\vec{a}$$

Inside $\oint \vec{A} \cdot d\vec{l} = A(2\cancel{\pi}r) = \int \vec{B} \cdot d\vec{a} = \mu_0 n I (\cancel{\pi} r^2)$ $\vec{A} = \frac{\mu_0 n I}{2} r \hat{\phi} \quad r < R$

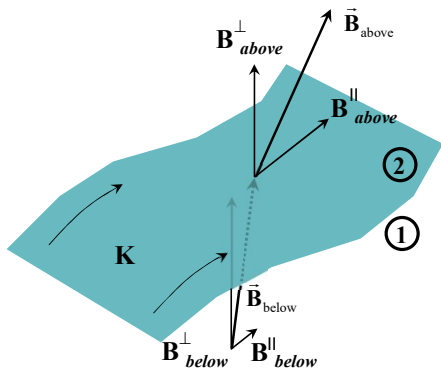
Outside $A(2\cancel{\pi}r) = \int \vec{B} \cdot d\vec{a} = \mu_0 n I (\cancel{\pi} R^2)$ $\vec{A} = \frac{\mu_0 n I}{2} \frac{R^2}{r} \hat{\phi} \quad r > R$

Verify by calculating curl $\vec{A} = \vec{B}$? div $\vec{A} = 0$?

Magnetostatic Boundary Conditions

Electric field suffers discontinuity at the surface charge

Magnetic field suffers discontinuity at the surface current



BC - Summary $\vec{\nabla} \cdot \vec{B} = 0$ $\vec{\nabla} \times \vec{B} = \mu_0 \mathbf{J}$

$$B_{\text{above}}^{\perp} = B_{\text{below}}^{\perp}$$

$$B_{\text{above}}^{\parallel} - B_{\text{below}}^{\parallel} = \mu_0 K$$

$$B_{\text{above}}^{\parallel} = B_{\text{below}}^{\parallel}$$



$$\vec{B}_{\text{above}} - \vec{B}_{\text{below}} = \mu_0 (\vec{K} \times \hat{n})$$

$$\vec{A}_{\text{above}} = \vec{A}_{\text{below}}$$

$$\frac{\partial \vec{A}_{\text{above}}}{\partial n} - \frac{\partial \vec{A}_{\text{below}}}{\partial n} = -\mu_0 \vec{K}$$

$$E_{\text{above}}^{\perp} - E_{\text{below}}^{\perp} = \frac{1}{\epsilon_0} \sigma$$

$$E_{\text{above}}^{\parallel} = E_{\text{below}}^{\parallel}$$

$$\vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0$$

$$\vec{\nabla} \times \vec{E} = 0$$



$$\vec{E}_{\text{above}} - \vec{E}_{\text{below}} = \frac{\sigma}{\epsilon_0} \hat{n}$$

$$V_{\text{above}} = V_{\text{below}}$$

$$\frac{\partial V_{\text{above}}}{\partial n} - \frac{\partial V_{\text{below}}}{\partial n} = -\frac{1}{\epsilon_0} \sigma$$

Magnetostatic Boundary Conditions

Electric field suffers discontinuity at the surface charge

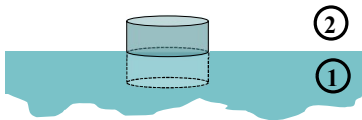
Magnetic field suffers discontinuity at the surface current



Magnetostatic Boundary Conditions

Electric field suffers discontinuity at the surface charge

Magnetic field suffers discontinuity at the surface current



Magnetostatic Boundary Conditions

Electric field suffers discontinuity at the surface charge

Magnetic field suffers discontinuity at the surface current

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\begin{aligned}\int_V (\vec{\nabla} \cdot \vec{B}) d\tau &= \oint \vec{B} \cdot d\vec{a} = \vec{B}_2 \cdot \hat{n}_2 \Delta a + \vec{B}_1 \cdot \hat{n}_1 \Delta a + w \\ &= \vec{B}_2 \cdot \hat{n} \Delta a - \vec{B}_1 \cdot \hat{n} \Delta a + w = \hat{n} \cdot (\vec{B}_2 - \vec{B}_1) \Delta a + w = 0\end{aligned}$$

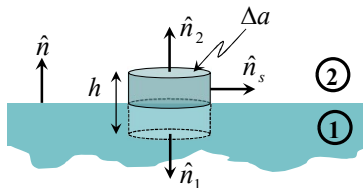
$$\lim_{h \rightarrow 0} \hat{n} \cdot (\vec{B}_2 - \vec{B}_1) \Delta a + \cancel{\lim_{h \rightarrow 0} w} = 0$$



$$\hat{n} \cdot (\vec{B}_2 - \vec{B}_1) = 0$$

$$B_{\text{above}}^{\perp} = B_{\text{below}}^{\perp}$$

Normal comp of B



Magnetostatic Boundary Conditions

Electric field suffers discontinuity at the surface charge

Magnetic field suffers discontinuity at the surface current

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} \quad \Rightarrow \quad \int_S (\vec{\nabla} \times \vec{B}) \cdot d\vec{a} = \mu_0 \int_S \vec{J} \cdot d\vec{a} \quad \Rightarrow \quad \oint_C \vec{B} \cdot d\vec{l} = \mu_0 \int_S \vec{J} \cdot d\vec{a}$$

$$\begin{aligned} \vec{B}_2 \cdot \hat{t}_2 \Delta l + \vec{B}_1 \cdot \hat{t}_1 \Delta l + w &= \vec{B}_2 \cdot \hat{t} \Delta l - \vec{B}_1 \cdot \hat{t} \Delta l + w = \hat{t} \cdot (\vec{B}_2 - \vec{B}_1) \Delta l + w \\ &= (\hat{n}' \times \hat{n}) \cdot (\vec{B}_2 - \vec{B}_1) \Delta l + w = \hat{n}' \cdot \hat{n} \times (\vec{B}_2 - \vec{B}_1) \Delta l + w = \mu_0 \vec{J} \cdot \hat{n}' h \Delta l \end{aligned}$$

$$\lim_{h \rightarrow 0} (\hat{n}' \cdot \hat{n} \times (\vec{B}_2 - \vec{B}_1) \Delta l + w) = \mu_0 \lim_{h \rightarrow 0} \vec{J} \cdot \hat{n}' h \Delta l$$

$$\hat{n}' \cdot [\hat{n} \times (\vec{B}_2 - \vec{B}_1)] = \mu_0 \hat{n}' \cdot \lim_{h \rightarrow 0} (h \vec{J}) = \mu_0 \hat{n}' \cdot \vec{K}$$

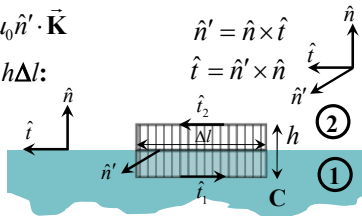
Total current ΔI through the area $h \Delta l$:

$$\Delta I = J h \Delta l$$

$$\lim_{h \rightarrow 0} \Delta I = \lim_{h \rightarrow 0} (J h \Delta l) = K \Delta l$$



$$\hat{n} \times (\vec{B}_2 - \vec{B}_1) = \mu_0 \vec{K}$$



Magnetostatic Boundary Conditions

Electric field suffers discontinuity at the surface charge

Magnetic field suffers discontinuity at the surface current

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} \quad \rightarrow \quad \hat{n} \times (\vec{B}_2 - \vec{B}_1) = \mu_0 \vec{K}$$

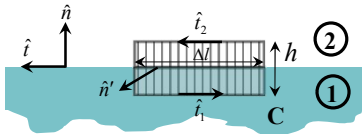
$$\hat{n} \times (\vec{B}_2^{\parallel} - \vec{B}_1^{\parallel}) = \mu_0 \vec{K}$$

$$\vec{B}_2^{\parallel} - \vec{B}_1^{\parallel} = \mu_0 (\vec{K} \times \hat{n})$$

$$\vec{B} = \vec{B}^{\perp} + \vec{B}^{\parallel} = B^{\perp} \hat{n} + \vec{B}^{\parallel}$$

$$\hat{n} \times \vec{B} = \hat{n} \times (B^{\perp} \hat{n} + \vec{B}^{\parallel}) = \hat{n} \times \vec{B}^{\parallel}$$

$$\begin{aligned} (\hat{n} \times \vec{B}) \times \hat{n} &= -\hat{n} \times (\hat{n} \times \vec{B}) \\ &= -[\hat{n}(\hat{n} \cdot \vec{B}) - \vec{B}(\hat{n} \cdot \hat{n})] \\ &= \vec{B} - B^{\perp} \hat{n} = \vec{B}^{\parallel} \end{aligned}$$



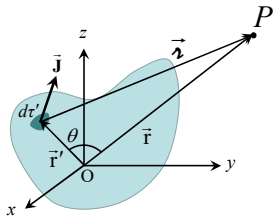
Multipole expansion of vector potential

$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\mathbf{r}')}{r} d\tau'$$

From cosine law:

$$\begin{aligned} r &= (r^2 + r'^2 - 2rr' \cos \theta)^{1/2} \\ &= r \left(1 + \frac{r'^2}{r^2} - \frac{2r'}{r} \cos \theta \right)^{1/2} \\ &\approx r \left(1 - \frac{2r'}{r} \cos \theta \right)^{1/2} \quad r \gg r' \end{aligned}$$

$$\begin{aligned} \frac{1}{r} &= \frac{1}{r} \left(1 - \frac{2r'}{r} \cos \theta \right)^{-1/2} \\ &\approx \frac{1}{r} \left(1 + \frac{r'}{r} \cos \theta \right) \end{aligned}$$

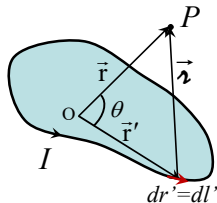


$$\begin{aligned} \vec{A}(\mathbf{r}) &= \frac{\mu_0}{4\pi r} \underbrace{\int \vec{J}(\mathbf{r}') d\tau'}_{\text{monopole}} \\ &+ \frac{\mu_0}{4\pi r^2} \underbrace{\int (\hat{\mathbf{r}} \cdot \mathbf{r}') \vec{J}(\mathbf{r}') d\tau'}_{\text{dipole}} + \dots \end{aligned}$$

Multipole expansion of vector potential

Current Carrying loop

$$\begin{aligned}\vec{\mathbf{A}}(\mathbf{r}) &= \frac{\mu_0}{4\pi} \oint \frac{\mathbf{I}}{r} d\vec{l}' \\ &= \underbrace{\frac{\mu_0 I}{4\pi r} \oint d\vec{l}'}_{\text{monopole}} + \underbrace{\frac{\mu_0 I}{4\pi r^2} \oint r' \cos \theta d\vec{l}'}_{\text{dipole}} \dots\end{aligned}$$



$$\begin{aligned}\vec{\mathbf{A}}_{\text{dip}}(\mathbf{r}) &= \frac{\mu_0 I}{4\pi r^2} \oint r' \cos \theta d\vec{l}' \\ &= \frac{\mu_0 I}{4\pi r^2} \oint (\mathbf{r}' \cdot \hat{\mathbf{r}}) d\vec{l}' \\ &= -\frac{\mu_0 I}{4\pi r^2} \hat{\mathbf{r}} \times \int d\vec{a}' = -\frac{\mu_0}{4\pi r^2} \hat{\mathbf{r}} \times \underbrace{I \int d\vec{a}'}_{\vec{\mathbf{m}}} = \frac{\mu_0}{4\pi} \frac{\vec{\mathbf{m}} \times \hat{\mathbf{r}}}{r^2}\end{aligned}$$

$$\frac{1}{r} = \frac{1}{r} \left(1 + \frac{r'}{r} \cos \theta \right)$$

$$\mathbf{r}' \cdot \hat{\mathbf{r}} = r' \cos \theta$$

$$\oint (\hat{\mathbf{r}} \cdot \mathbf{r}') dl = -\hat{\mathbf{r}} \times \int d\vec{a}' \quad \text{Eq. 1.108}$$

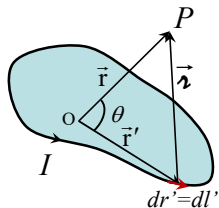
$\vec{\mathbf{m}}$

Multipole expansion of vector potential

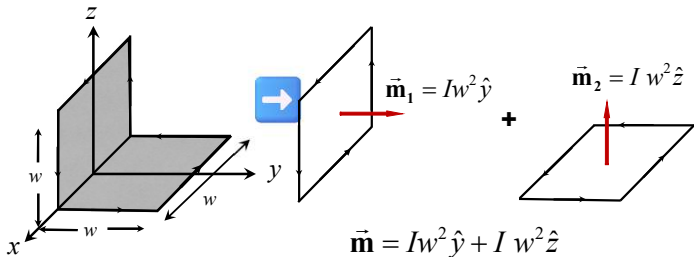
Current Carrying loop

$$\vec{A}_{\text{dip}}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \hat{\mathbf{r}}}{r^2}$$

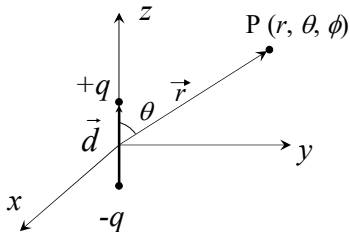
$$\vec{m} = I \int d\vec{a}' = I \vec{a}$$



Example:



Electric Dipole



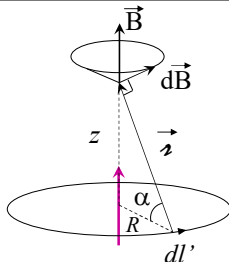
Potential

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{q\vec{d} \cdot \hat{r}}{r^2}$$

Electric Field

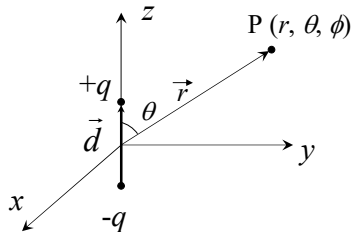
$$\vec{E}_{dip}(r, \theta) = \frac{1}{4\pi\epsilon_0} \frac{p}{r^3} \left[(2 \cos \theta) \hat{r} + (\sin \theta) \hat{\theta} \right]$$

Magnetic Dipole



$$\vec{B}(z) = \frac{\mu_0}{2\pi} \underbrace{(I \pi R^2)}_{\vec{m}} \frac{1}{z^3} \hat{z}$$

Electric Dipole



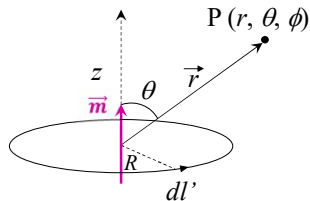
Potential

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{q\vec{d} \cdot \hat{r}}{r^2}$$

Electric Field

$$\vec{E}_{dip}(r, \theta) = \frac{1}{4\pi\epsilon_0} \frac{p}{r^3} \left[(2 \cos \theta) \hat{r} + (\sin \theta) \hat{\theta} \right]$$

Magnetic Dipole



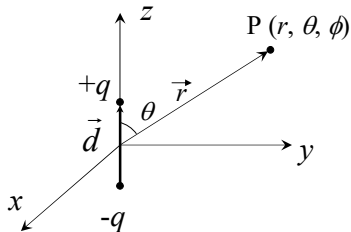
Potential

$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int I \frac{d\mathbf{l}'}{r}$$

$$\vec{A}_{dip}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \hat{r}}{r^2}$$

$$\vec{m} = I \int d\vec{a} = I \vec{a}$$

Electric Dipole



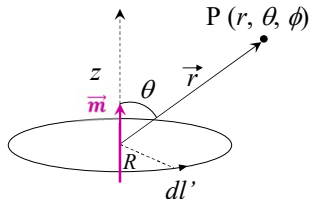
Potential

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{q\vec{d} \cdot \hat{r}}{r^2}$$

Electric Field

$$\vec{E}_{dip}(r, \theta) = \frac{1}{4\pi\epsilon_0} \frac{p}{r^3} \left[(2 \cos \theta) \hat{r} + (\sin \theta) \hat{\theta} \right]$$

Magnetic Dipole



Potential

$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int I \frac{d\mathbf{l}'}{r}$$

$$\vec{A}_{dip}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \hat{r}}{r^2}$$

$$\vec{m} = I \int d\vec{a} = I \vec{a}$$

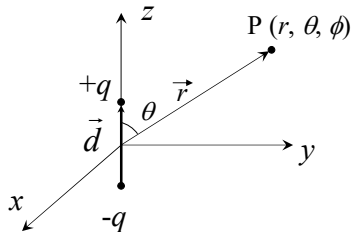
Magnetic Field

$$\vec{B}_{dip} = \vec{\nabla} \times \vec{A}$$

$$\vec{B}_{dip}(r, \theta) = \frac{\mu_0 m}{4\pi r^3} \left[(2 \cos \theta) \hat{r} + (\sin \theta) \hat{\theta} \right]$$

Dipole field

Electric Dipole



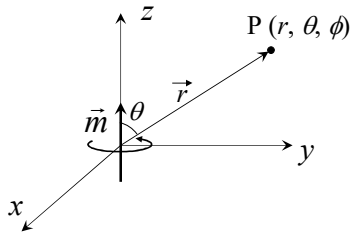
Potential

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{q\vec{d} \cdot \hat{r}}{r^2}$$

Electric Field

$$\vec{E}_{dip}(r, \theta) = \frac{1}{4\pi\epsilon_0} \frac{p}{r^3} \left[(2 \cos \theta) \hat{r} + (\sin \theta) \hat{\theta} \right]$$

Magnetic Dipole



Potential

$$\vec{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int I \frac{d\mathbf{l}'}{r}$$

$$\vec{A}_{dip}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \hat{r}}{r^2}$$

$$\vec{m} = I \int d\vec{a} = I \vec{a}$$

Magnetic Field

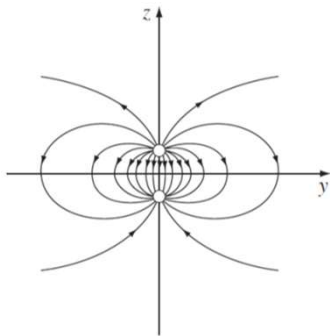
$$\vec{B}_{dip} = \vec{\nabla} \times \vec{A}$$

$$\vec{B}_{dip}(r, \theta) = \frac{\mu_0 m}{4\pi r^3} \left[(2 \cos \theta) \hat{r} + (\sin \theta) \hat{\theta} \right]$$

Dipole field

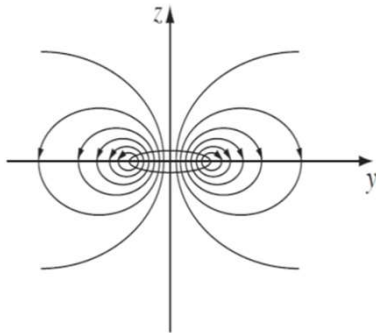
Electric Dipole

Physical



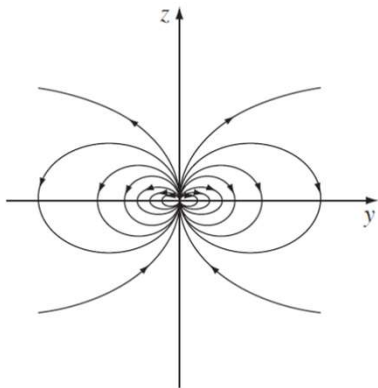
Magnetic Dipole

Physical



Electric Dipole

Pure



Magnetic Dipole

Pure

